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ShanghaiTech University

Electric Circuits(EE111)

Tutorial 2

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Instructor: Chaofeng Ye

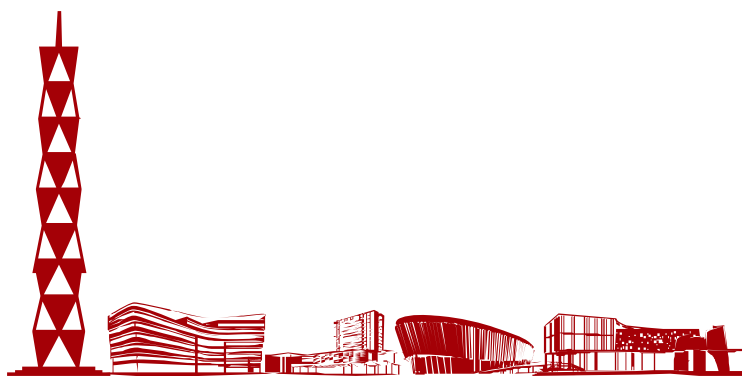


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Circuit Theorems



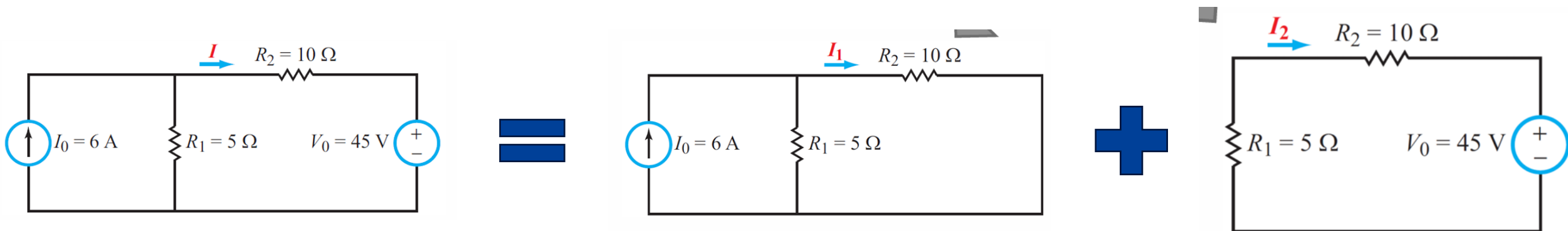
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Superposition 叠加定理

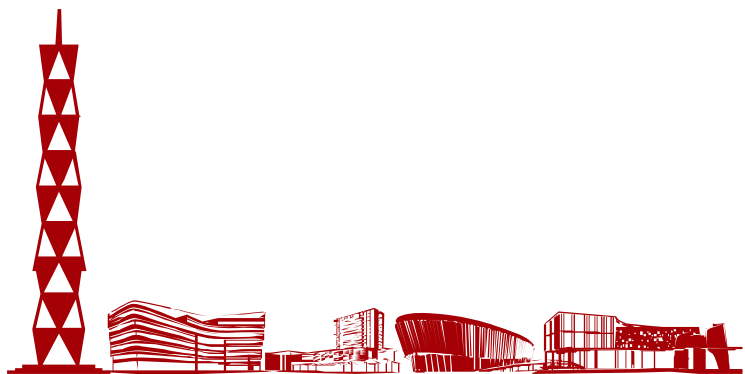


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在线性电路中，一个元件上的电压（或电流）是由于每个独立电源单独作用时，该元件上的电压（或电流）的代数和。



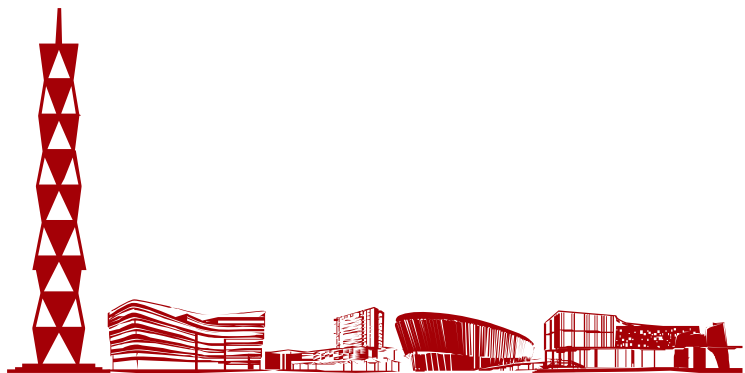
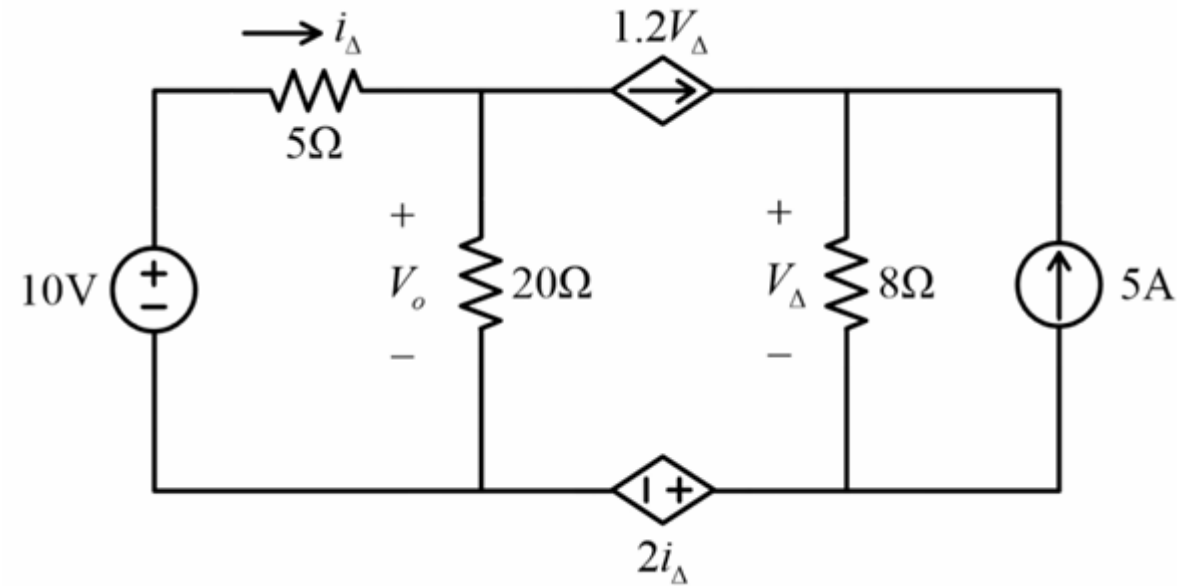
- 单独作用时，其余电源置零
- 叠加时注意电压/电流的方向
- 受控源在任何情况下都应保留
- 不能用叠加定理产生的分量计算功率



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1. Use the principle of superposition to find V_o in the circuit.



- Thevenin' s theorem
- Source transformation
- Norton' s theorem



等效电源

Method 2: Open/short circuit

1. Analyze circuit to find v_{oc}
2. Analyze circuit to find i_{sc}

$$v_{Th} = v_{oc}$$

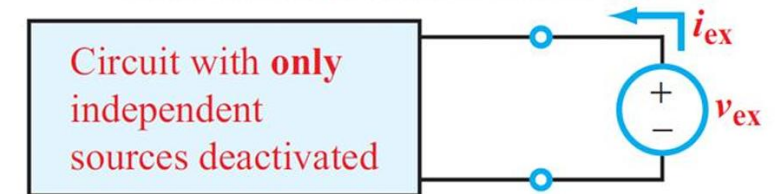
$$R_{Th} = \frac{v_{Th}}{i_{sc}}$$

Method 3:

Method 1: Equivalent Resistance

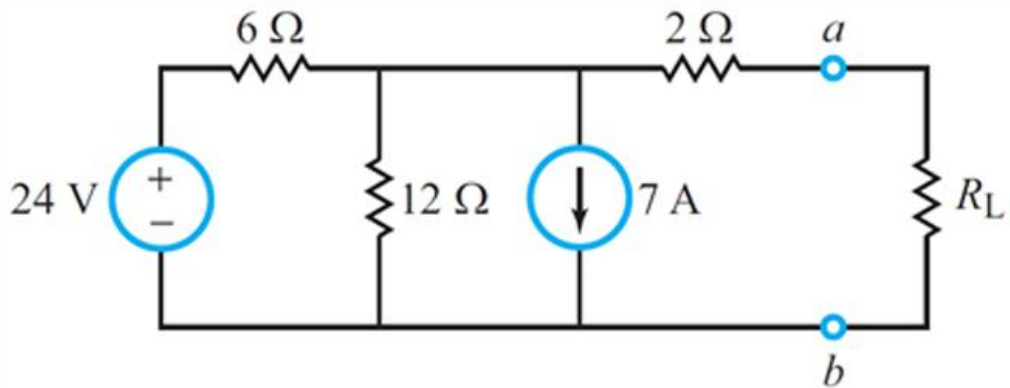
1. Analyze circuit to find v_{oc}
2. **Deactivate** all **independent** sources by replacing voltage sources with short circuits and current sources with open circuits.
3. Simplify circuit to find equivalent resistance.

External-Source Method



If a circuit contains both dependent and independent sources, R_{Th} can be determined by (a) deactivating independent sources (only), (b) adding an external source v_{ex} , and then (c) solving the circuit to determine i_{ex} . The solution is $R_{Th} = v_{ex}/i_{ex}$.

Still $v_{Th} = v_{oc}$



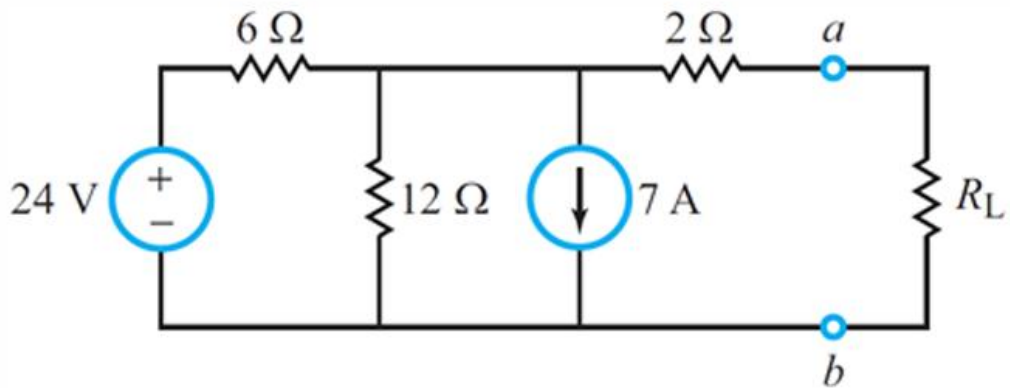
Method 1:



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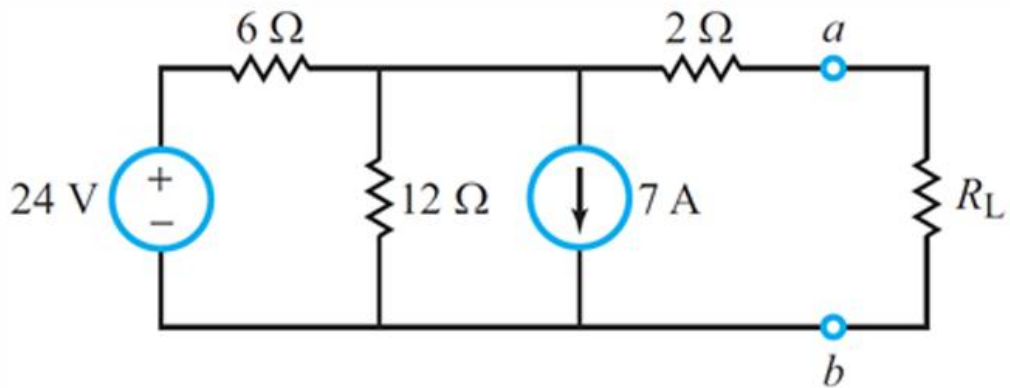
Method 2:



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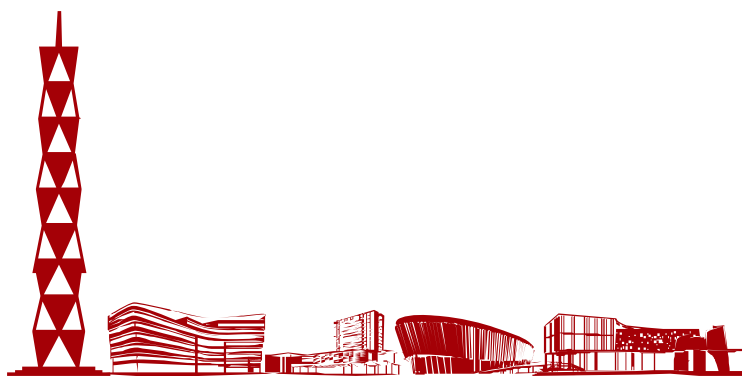
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Method 3:



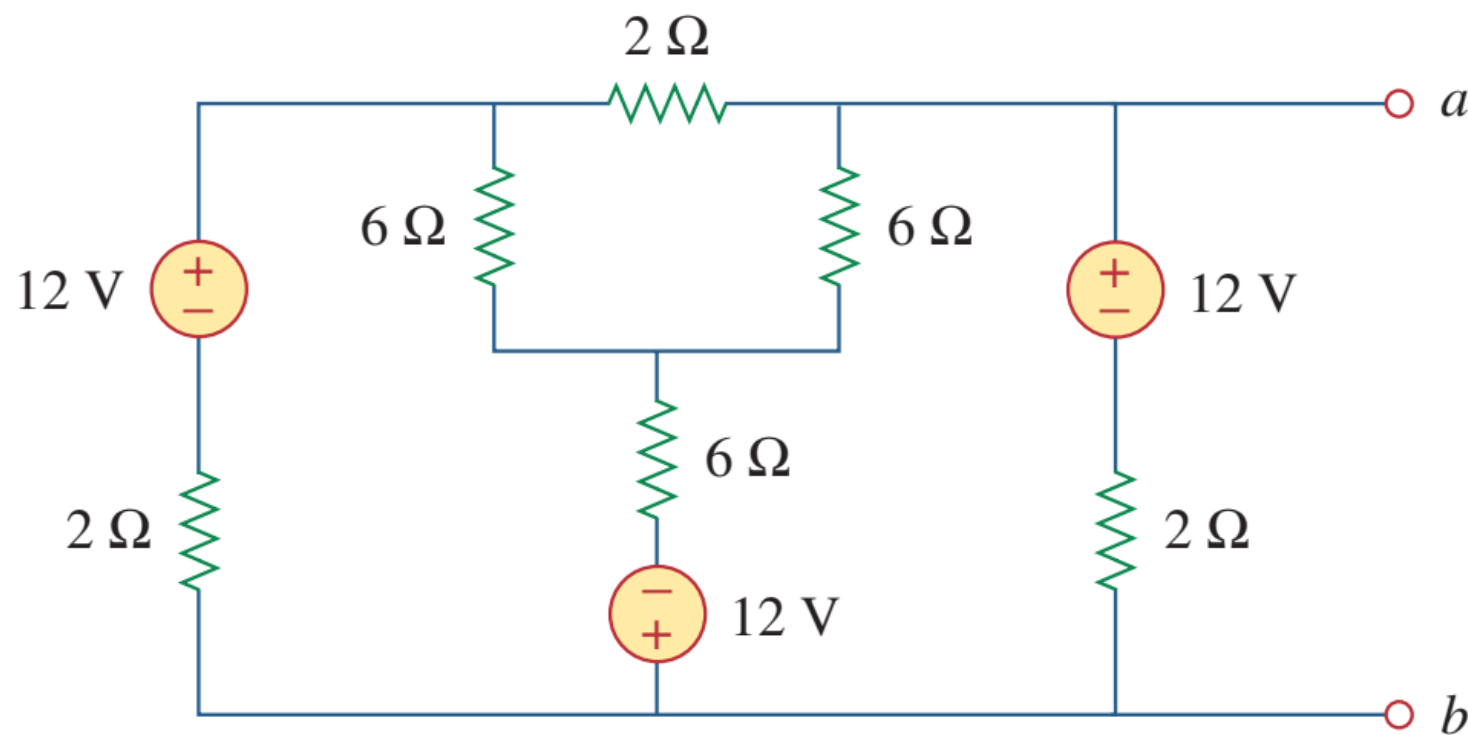
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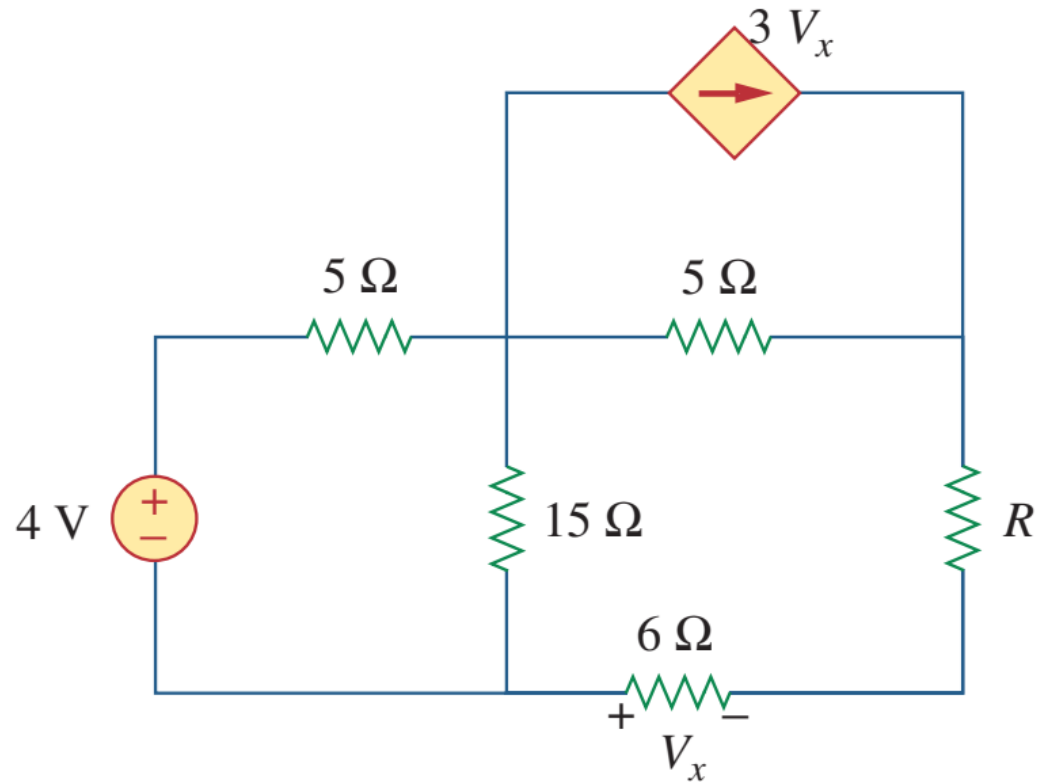
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2. Obtain the Thevenin and Norton equivalent circuits at terminals of the circuit

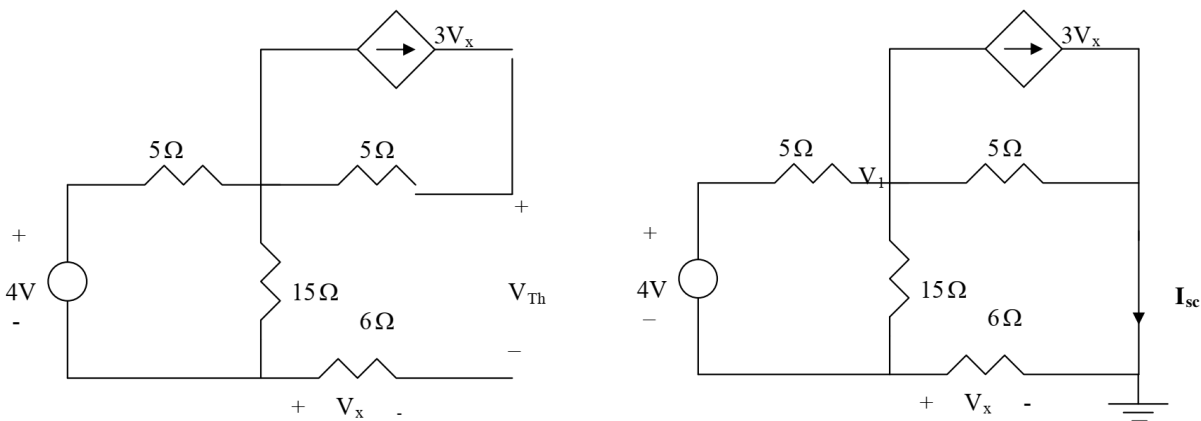


3. Determine the maximum power delivered to the variable resistor R shown in the circuit





We find the Thevenin equivalent across the 10-ohm resistor. To find V_{Th} , consider the circuit below.



From the figure,

$$V_x = 0, \quad V_{Th} = \frac{15}{15+5}(4) = 3V$$

To find R_{eq} , consider the circuit below:

At node 1,

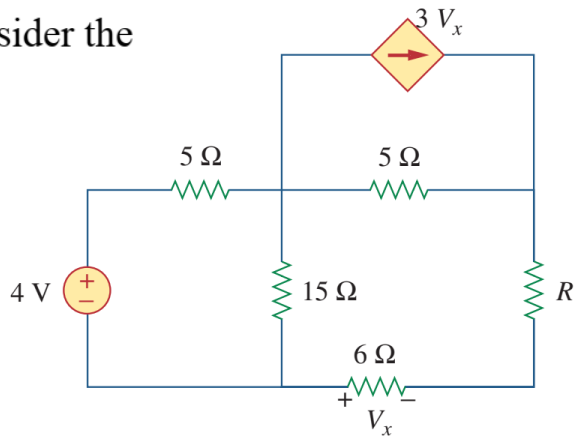
$$[(V_1 - V_x)/15] + [(V_1 - (4 + V_x))/5] + [(V_1 - 0)/5] + 3V_x = 0 \text{ or}$$

$$0.4667V_1 + 2.733V_x = 0.8 \quad (1)$$

At node x,

$$[(V_x - 0)/6] + [((V_x + 4) - V_1)/5] + [(V_x - V_1)/15] = 0 \text{ or}$$

$$-(0.2667)V_1 + 0.4333V_x = -0.8 \quad (2)$$



Adding (1) and (2) together lead to,

$$(0.4667 - 0.2667)V_1 + (2.733 + 0.4333)V_x = 0 \text{ or } V_1 = -(3.166/0.2)V_x = -15.83V_x$$

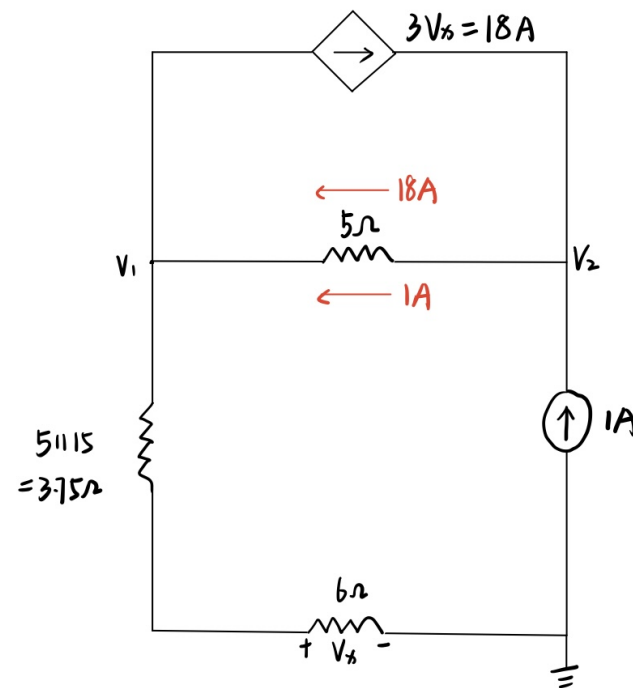
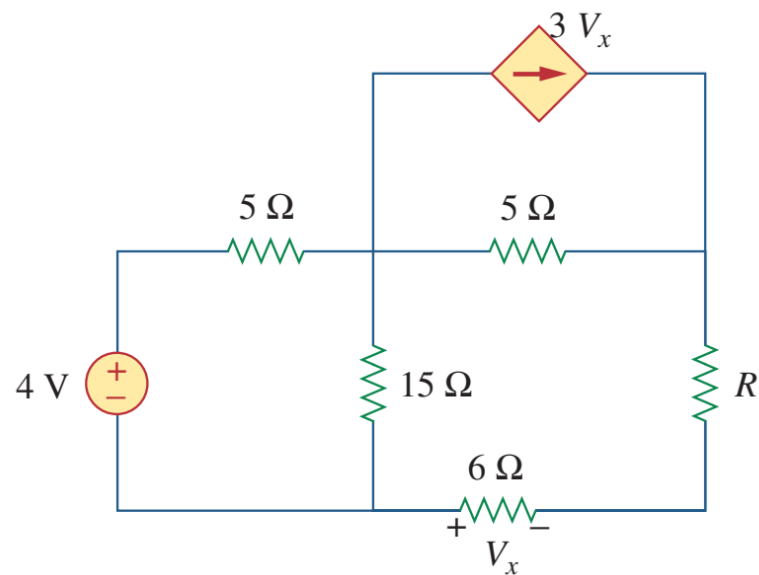
Now we can put this into (1) and we get,

$$0.4667(-15.83V_x) + 2.733V_x = 0.8 = (-7.388 + 2.733)V_x = -4.655V_x \text{ or } V_x = -0.17186V$$

$$I_{sc} = -V_x/6 = 0.02864 \text{ and } R_{eq} = 3/(0.02864) = 104.75 \Omega$$

Clearly setting the load resistance to 104.75Ω means that the circuit will deliver maximum power to it. Therefore,

$$p_{max} = [3/(2 \times 104.75)]^2 \times 104.75 = \mathbf{21.48 \text{ mW}}$$



$$V_x = 6V$$

$$V_1 = 16 + 3.75 \times 1 = 9.75V$$

$$V_2 = 19 \times 5 + V_1 = 104.75V$$

$$R_{eq} = \frac{V_2 - 0}{1} = 104.75\Omega$$

op-amp

An **ideal op amp** is an amplifier with infinite open-loop gain, infinite input resistance, and zero output resistance.



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Although assuming an ideal op amp provides only an approximate analysis, most modern amplifiers have such large gains and input impedances that the approximate analysis is a good one. Unless stated otherwise, we will assume from now on that every op amp is ideal.

For circuit analysis, the ideal op amp is illustrated in Fig. 5.8, which is derived from the nonideal model in Fig. 5.4. Two important characteristics of the ideal op amp are:

1. The currents into both input terminals are zero:

$$i_1 = 0, \quad i_2 = 0 \quad (5.5)$$

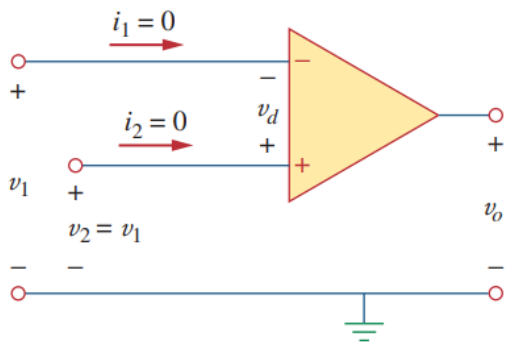
This is due to infinite input resistance. An infinite resistance between the input terminals implies that an open circuit exists there and current cannot enter the op amp. But the output current is not necessarily zero according to Eq. (5.1).

2. The voltage across the input terminals is equal to zero; i.e.,

$$v_d = v_2 - v_1 = 0 \quad (5.6)$$

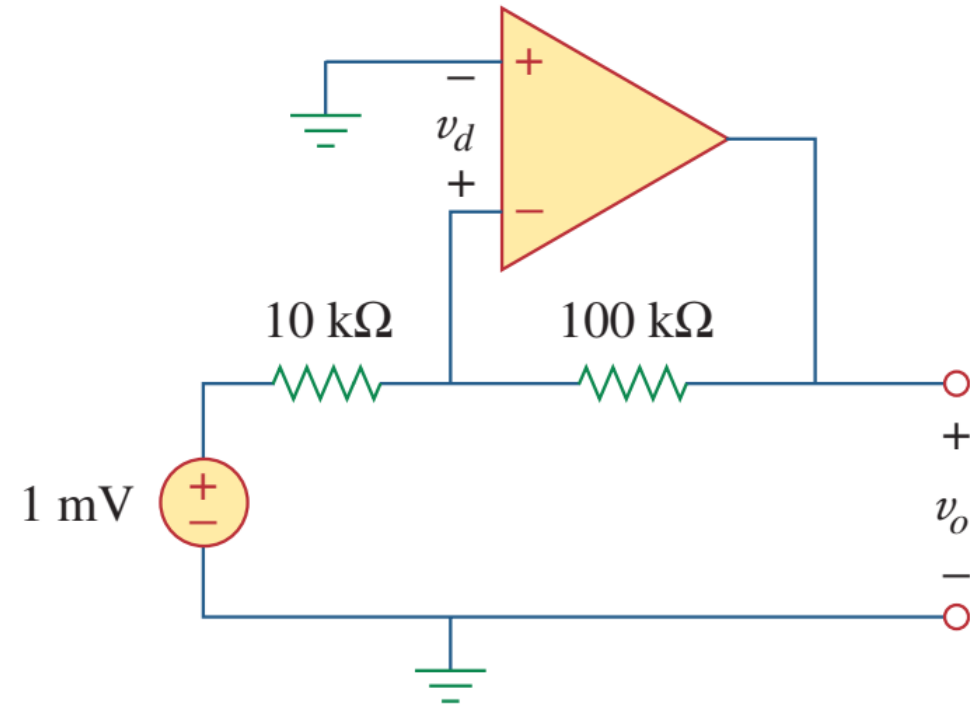
or

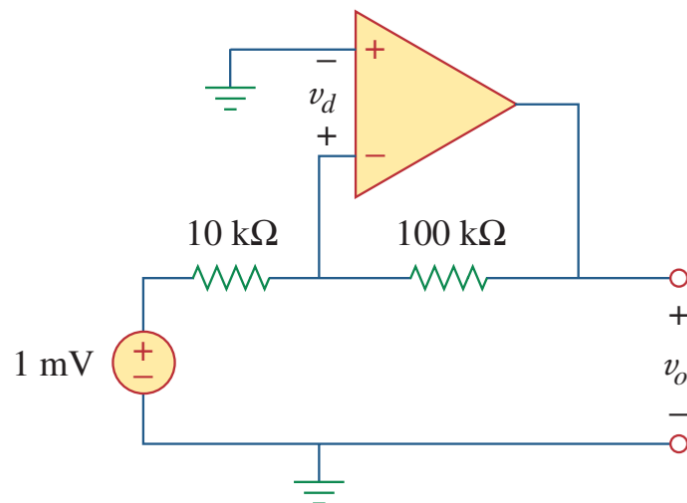
$$v_1 = v_2 \quad (5.7)$$





4. The op amp has $R_i = 100 \text{ k}\Omega$,
 $R_o = 100 \text{ }\Omega$, $A = 100,000$. Find the
differential voltage v_d and the output
voltage v_o .





At node 1, $(V_S - V_1)/10 \text{ k} = [V_1/100 \text{ k}] + [(V_1 - V_0)/100 \text{ k}]$

$$10 V_S - 10 V_1 = V_1 + V_1 - V_0$$

which leads to $V_1 = (10V_S + V_0)/12$

At node 2, $(V_1 - V_0)/100 \text{ k} = (V_0 - (-AV_d))/100$

But $V_d = V_1$ and $A = 100,000$,

$$V_1 - V_0 = 1000 (V_0 + 100,000V_1)$$

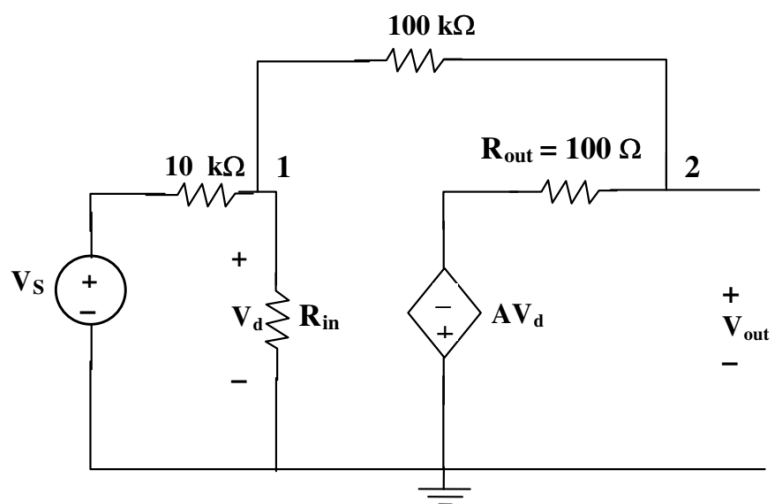
$$0 = 1001V_0 + 99,999,999[(10V_S + V_0)/12]$$

$$0 = 83,333,332.5 V_S + 8,334,334.25 V_0$$

which gives us $(V_0/V_S) = -10$ (for all practical purposes)

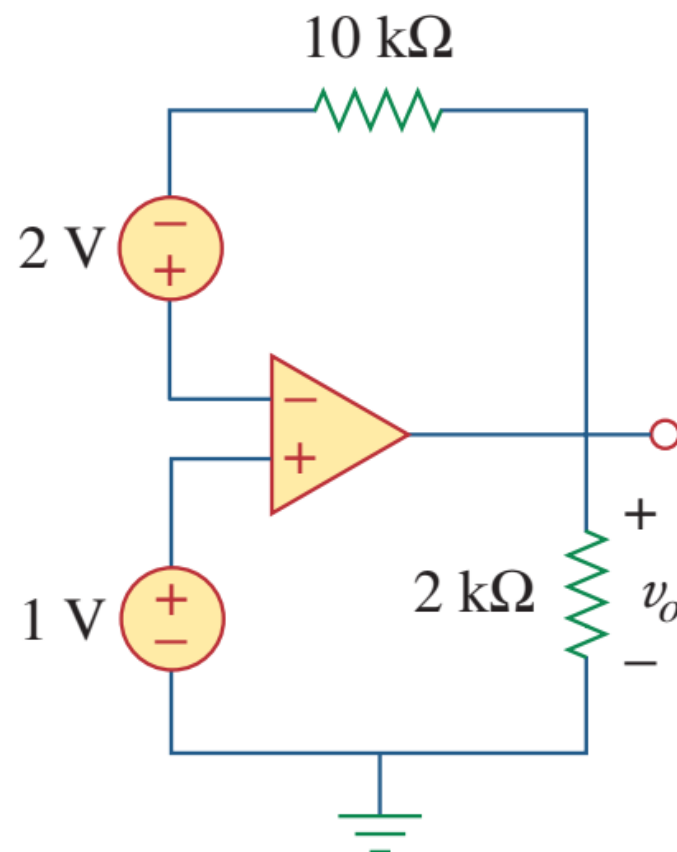
If $V_S = 1 \text{ mV}$, then $V_0 = -10 \text{ mV}$

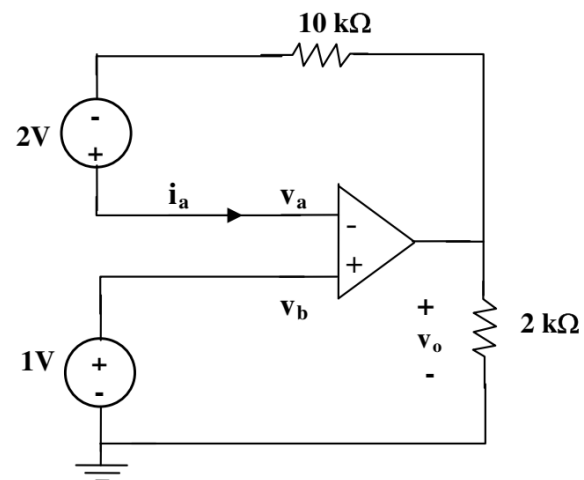
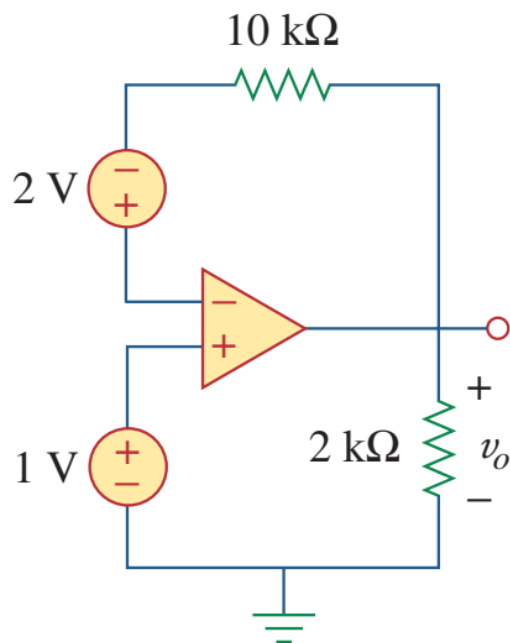
Since $V_0 = A V_d = 100,000 V_d$, then $V_d = (V_0/10^5) \text{ V} = -100 \text{ nV}$



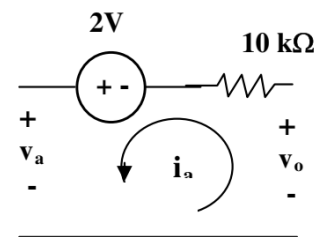


5. Obtain v_o for the op amp circuit

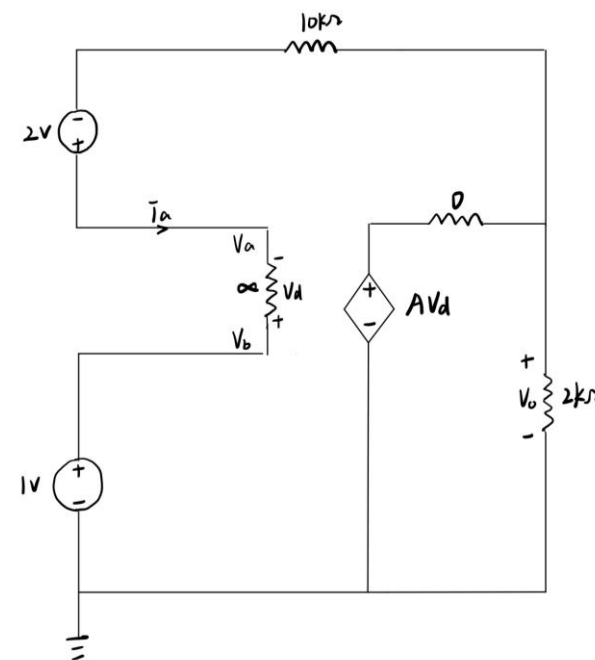




(a)



(b)



Since $v_a = v_b = 1\text{V}$ and $i_a = 0$, no current flows through the $10\text{ k}\Omega$ resistor. From Fig. (b),

$$-v_a + 2 + v_o = 0 \longrightarrow v_o = v_a - 2 = 1 - 2 = -1\text{V}$$

6. Obtain i_o in the instrumentation amplifier circuit of Fig. 5.27.

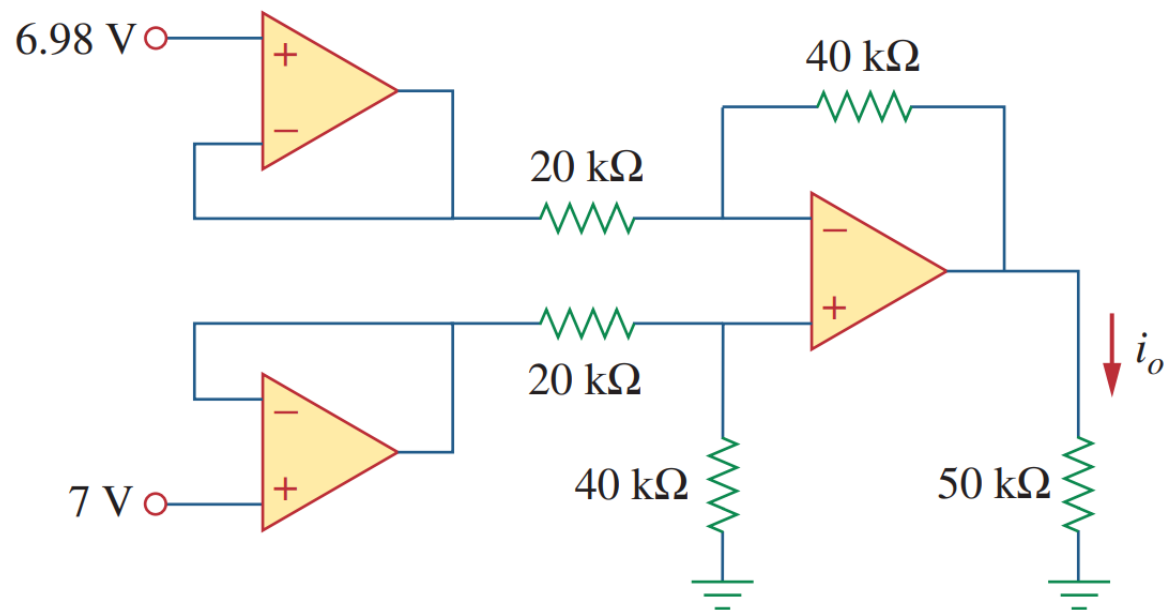
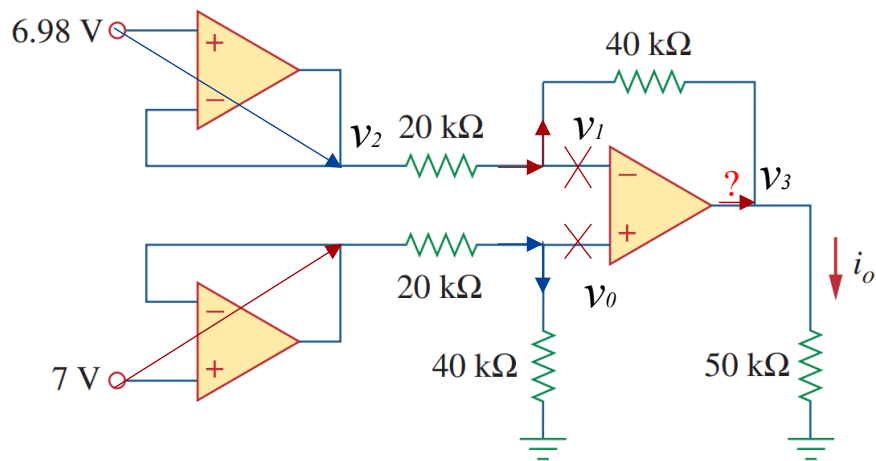


Figure 5.27

Instrumentation amplifier; for Practice Prob. 5.8.



$$\frac{7 - v_0}{20000} = \frac{v_0 - 0}{40000}$$

$$v_0 = v_1$$

$$\frac{6.98 - v_1}{20000} = \frac{v_1 - v_3}{40000}$$

$$i_o = \frac{V_3}{50000} = -800\mu A$$

A red-tinted photograph of a modern architectural walkway. The walkway features a series of tall, cylindrical columns supporting a curved, overhanging roof structure. The ground is a smooth, reflective surface, likely polished concrete or stone, which mirrors the columns and the sky. In the background, a tall, slender lattice tower stands prominently against a hazy sky. The overall scene is a blend of modern design and urban landscape.

Thanks for your attention!